1. NAME OF THE MEDICINAL PRODUCT

HIDRASEC 100 mg hard capsules

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each capsule contains 100 mg of racecadotril.
Excipients with known effects:
Each capsule contains 41 mg lactose monohydrate.

For a full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Hard capsule.
Ivory coloured capsules.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

HIDRASEC is indicated for the symptomatic treatment of acute diarrhoea in adults when causal treatment is not possible.

If causal treatment is possible, racecadotril can be administered as a complementary treatment.

4.2 Posology and method of administration

For oral use.

<u>Adults</u>:
One capsule initially regardless of the time of day. Then, one capsule three times daily preferably before the main meals. Treatment should be continued until two normal stools are recorded.
Treatment should not exceed 7 days.
Long-term treatment with racecadotril is not recommended.

<u>Special populations</u>:
Children: There are specific formulations intended for infants, children and adolescents.
Older people: Dosage adjustment is not necessary in the older people (see section 5.2).
Caution is advised in patients with hepatic or renal impairment.

4.3 Contraindications

- Hypersensitivity to the active substance or to any of the excipients.
- Patients who have reported angioedema with angiotensin converting enzyme inhibitors (such as captopril, enalapril, Lisinopril, perindopril, Ramipril) should not take racecadotril.

4.4 Special warnings and precautions for use

<u>Precautions for use</u>:
The administration of racecadotril does not modify the usual rehydration regimens.

The presence of bloody or purulent stools and fever may indicate the presence of invasive bacteria as a reason for diarrhoea, or the presence of other severe disease, warranting causal (e.g antibiotic) treatment or further investigation. Therefore, racecadotril should not be administered under these conditions.

Racecadotril may be given concomitantly with antibiotics in case of acute diarrhoea with a bacterial cause as a complementary treatment.
Use of racecadotril in antibiotic-associated diarrhoea and chronic diarrhoea is not recommended due to insufficient data.

There are limited data in patients with renal or hepatic impairment. These patients should be treated with caution (see section 5.2).

There is a possible reduced availability in patients with prolonged vomiting.

Warning:
This medicinal product contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Occurrence of skin reactions has been reported with the use of the product. These are in most cases mild and do not require treatment but in some cases they can be severe, even life-threatening.

Association with racecadotril cannot be fully excluded. When experiencing severe skin reactions, the treatment has to be stopped immediately.

Hypersensitivity/Angioedema have been reported in patients with racecadotril. This may occur at any time during therapy. Patients with a history of angioedema unrelated to racecadotril therapy may be at increased risk of angioedema.

Severe cutaneous adverse reactions (SCARs):
Severe cutaneous adverse reactions (SCARs) including drug reaction with eosinophilia and systemic symptoms (DRESS), which can be life-threatening or fatal, have been reported in association with racecadotril treatment. Patients should be advised of the signs and symptoms and monitored closely for skin reactions. If signs and symptoms suggestive of DRESS appear, racecadotril should be withdrawn immediately and an alternative treatment considered. If the patient has developed DRESS with the use of racecadotril, treatment with racecadotril must not be restarted in these patients at any time.

4.5 Interaction with other medicinal products and other forms of interaction

Angiotensin converting enzyme inhibitors (such as captopril, enalapril, lisinopril, fosinopril, perindopril, ramipril) are known to induce angioedema. This risk could be increased in presence of racecadotril.
In humans, concomitant treatment with racecadotril and loperamide or nifuroxazide does not modify the kinetics of racecadotril.

4.6 Fertility, pregnancy and lactation

Pregnancy:
There are no adequate data from the use of racecadotril in pregnant women. Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryo-foetal development, parturition or postnatal development. However, since no specific clinical studies are available, racecadotril should not be administered to pregnant women.

Lactation:
Due to the lack of information on secretion of Hidrasec in human milk, the product should not be administered to breastfeeding women.

4.7 Effects on ability to drive and use machines

Racecadotril has no or negligible influence on the ability to drive and use machines.

4.8 Undesirable effects

The following adverse drug reactions listed below have occurred with racecadotril more often than with placebo, or have been reported during post-marketing surveillance.

The frequency of adverse reactions is defined using the following convention: very common (≥ 1/10), common (≥ 1/100 to < 1/10), uncommon (≥ 1/1,000 to < 1/100), rare (≥ 1/10,000 to < 1/1,000), very rare (< 1/10,000), not known (cannot be estimated from the available data).

Severe cutaneous adverse reactions (SCARs) including drug reaction with eosinophilia and systemic symptoms (DRESS) have been reported in association with racecadotril treatment (see section 4.4).

Nervous system disorders

- Common: headache.

Skin and subcutaneous tissue disorders

- Uncommon: rash, erythema.
- Unknown: erythema multiforme, tongue oedema, face oedema, lip oedema, eyelid oedema, angioedema, urticaria, erythema nodosum, rash papular, prurigo, pruritus, toxic skin eruption, drug reaction with eosinophilia and systemic symptoms (DRESS).

Immune system disorders:
- Unknown: anaphylactic shock

Paediatric population
In infants or children treated with racecadotril the occurrence of tonsillitis has been reported as an uncommon adverse event. There are specific formulations for these age groups.

Serious skin reactions (including angioedema) have been reported in patients on racecadotril therapy. The incidence of these reactions is unknown but if they occur, racecadotril therapy must be discontinued and appropriate alternative therapy instituted. Patients should be aware not to take racecadotril again in these cases.

4.9 Overdose
So far sporadic cases of overdose without adverse events have been reported. In adults, single doses above 2 g, which is equivalent to 20 times the therapeutic dose, have been administered, and no harmful effects have been described.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Other antidiarrhoeals.

ATC code: A07XA04

Racecadotril is a prodrug that needs to be hydrolysed to its active metabolite thiorphan, which is an inhibitor of enkephalinase, a cell membrane peptidase located in various tissues, notably the epithelium of the small intestine. This enzyme contributes both to the hydrolysis of exogenous peptides and to the breakdown of endogenous peptides such as enkephalins. Consequently, racecadotril protects endogenous enkephalins that are physiologically active at digestive tract level, prolonging their antisecretory effect.

Racecadotril is a pure intestinal antisecretory active substance. It decreases the intestinal hypersecretion of water and electrolytes induced by cholera toxin or inflammation, and does not have effects on basal secretory activity. Racecadotril exerts rapid antidiarrhoeal action, without modifying the duration of intestinal transit.

Racecadotril does not produce abdominal distension. During its clinical development, racecadotril produced secondary constipation at a rate comparable to placebo.

When administered via the oral route, its activity is exclusively peripheral, with no effects on the central nervous system.

5.2 Pharmacokinetic properties

Absorption:
Following oral administration, racecadotril is rapidly absorbed. The initial time to plasma enkephalinase inhibition is 30 minutes.

The bioavailability of racecadotril is not modified by food, but peak activity is delayed by about one hour and a half.

Distribution:
In plasma, after oral administration of 14C-labeled racecadotril, measured exposure of radiocarbon was many orders of magnitude higher than in blood cells and 3-fold higher in in whole blood. Thus, the drug did not bind to blood cells to any significant extent. Radiocarbon distribution in other body tissues was moderate, as indicated by the mean apparent volume of distribution in plasma of
66.4 kg. Ninety percent of the active metabolite of racecadotril, thiorphan = (RS)-N-(1-oxo-2-(mercaptomethyl)-3-phenylpropyl) glycine, is bound to plasma proteins, mainly to albumin.

The pharmacokinetic properties of racecadotril are not modified as a result of repeat dosing or administration to elderly persons.

The duration and extent of the effect of racecadotril are dose-dependent.

In adults, time to peak plasma enkephalinase inhibition is approximately 2 hours and corresponds to 75% inhibition with the dose of 100 mg.

The duration of plasma enkephalinase inhibition is about 8 hours.

Metabolism:
The biological half-life of racecadotril, measured as plasma enkephalinase inhibition, is 3 hours approximately.

Racecadotril is rapidly hydrolysed to thiorphan, the active metabolite, which is in turn transformed into inactive metabolites.

Repeated administration of racecadotril does not cause any accumulation in the body.

In vitro data indicate that racecadotril/thiorphan and the four major inactive metabolites do not inhibit the major CYP enzymes isoforms 3A4, 2D6, 2C9, 1A2 and 2C19 to an extent that would be clinically relevant.

In vitro data indicate that racecadotril/thiorphan and the four major inactive metabolites do not induce the CYP enzymes isoforms (3A family, 2A6, 2B6, 2C9/2C19, 1A family, 2E1) and UGTs conjugating enzymes to an extent that would be clinically relevant.

Racecadotril does not modify protein binding of active substances strongly bound to proteins, such as tolbutamide, warfarin, niflumic acid, digoxin or phenytoin.

In patients with liver failure [cirrhosis, grade B of the Child-Pugh classification], the kinetic profile of the active metabolite of racecadotril showed similar Tmax and T½ and lesser Cmax (-65%) and AUC (- 29%) as compared to healthy subjects.
In patients with severe renal failure (creatinine clearance 11-39 ml/min), the kinetic profile of the active metabolite of racecadotril showed smaller Cmax (-49%) and greater AUC (+16%) and T½ as compared to healthy volunteers (creatinine clearance >70 ml/min).

In the paediatric population, pharmacokinetic results are similar to those of the adult population, reaching Cmax at 2 hours 30 min after administration. There is no accumulation after multiple dose administrated every 8 hours, for 7 days.

Excretion:
Racecadotril is eliminated as active and inactive metabolites. Elimination is mainly via the renal route, and to a much lesser extent via the faecal route. The pulmonary route is not significant.

5.3 Preclinical safety data

Chronic exposure in monkeys at a dose of 500 mg/kg/day resulted in generalized infections and reduced antibody responses to vaccination but no infection/immune depression was observed at 120 mg/kg/day. The clinical relevance of this finding is unknown.
Racecadotril was not immunotoxic in mice when given for up to 1 month.

Four-week toxicity studies in monkeys and dogs, relevant for the duration of treatment in human, do not point out any effect at doses up to 1250 mg/kg/day and 200 mg/kg, respectively.

Preclinical effects (e.g., severe, most likely aplastic anaemia, increased diuresis, ketonuria, diarrhoea) were observed only at exposures considered sufficiently in excess of the maximum human exposure. The clinical relevance is unknown.

No mutagenic or clastogenic effect of racecadotril has been found in the standard *in vitro* and *in vivo* tests.

Reproductive and developmental toxicity studies have not revealed any special effects of racecadotril.

A toxicity study in juvenile rats has not revealed any significant effects of racecadotril up to a dose of 160 mg/kg/day which is 35 times higher than the usual paediatric regimen (i.e. 4.5 mg/kg/day).

In animals, racecadotril reinforced the effects of butylhyoscine upon bowel transit and on the anticonvulsive effects of phenytoin.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Powder
Lactose monohydrate
Pregelatinised starch (maize)
Magnesium stearate
Silica Colloidal anhydrous

Capsule
Yellow iron oxide (E172)
Titanium dioxide (E171)
Gelatin

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

3 years.

6.4 Special precautions for storage

This medicinal product does not require any special storage conditions. Do not store above 30ºC. Keep out of reach of children.

6.5 Nature and contents of container

PVC-PVDC/ Aluminium blister.
Packs containing 6, 10, 20, 100 and 500 capsules.
Not all pack sizes may be marketed

6.6 Special precautions for disposal and other handling

No special requirements for disposal.

7. MANUFACTURER

Laboratoires Sophartex
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8. DATE OF REVISION OF THE TEXT

August 2024
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